Read the Instrument

Where the digits in a measurement come from, and how the glass in your hand decides how many of them you are allowed to write.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/read-the-instrument/

1 Before you open anything

Answer from what you already know. Nothing here is graded on being right — the point is to have a committed answer to compare against later.

PREDICTION

A 100 mL graduated cylinder is marked every 1 mL. The bottom of the meniscus sits a little below halfway between the 28 and the 29 mark.

a. Write the volume exactly as you would record it in a lab notebook.

____

b. Someone photographs that same tube and enlarges the picture until the marks are three times further apart on the screen. Write the volume you would record from the enlarged photograph.

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In one sentence: what decided how many digits you wrote each time?

____

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2 Read three levels off the glass

Open the demonstration. The picture is a close-up of one slice of the scale; the thin bar on the far left is the whole instrument, with a box showing which slice you are seeing.

1. In the **Instrument** panel, leave the tab on **Cylinder**.
2. Drag **Graduation spacing** until the value beside it reads **1 mL** (the middle position of that slider).
3. From the **Facts** panel, copy down the fixed values once:

Readout	Value
Marks every	
In view	
Instrument range	
Bottom-to-top of the curve	

1. Read the **bottom** of the curved surface, type the number into **Your reading**, and press **Check**.
2. Fill in one row below. The value labelled **correct reading** is drawn on the picture next to a dashed line.
3. Press **New reading** and repeat twice more.

Try	Your reading (mL)	“correct reading” (mL)	Digits after the decimal point	Significant figures
1				
2				
3				

a. The small grey line at the bottom of the feedback box restates the rule using the actual place names for this instrument. Copy it out.

b. In your three readings, which digits did the printed marks give you, and which digit did you supply? Which one carries the uncertainty?

> If the demonstration accepted a reading one step away from the one it drew, that is not a mistake being excused. The last digit is the estimated one, so ±1 in that place is the honest spread between two careful people reading the same glassware.

3 Change the marks, not the liquid

For this whole part, do *not* press **New reading**. Moving the **Graduation spacing** slider swaps you onto a different piece of glassware but keeps the liquid at the same height in the tube, so the only thing that changes is the scale ruled on the glass.

1. Stay on **Cylinder**. Drag **Graduation spacing** fully **left** — the value should read **10 mL**.
2. Record **Instrument range** from the Facts panel. Then read the level, type it, press **Check**, and record the **correct reading** the demonstration draws.
3. Move the slider one step right (**1 mL**) and repeat. Then one more step right (**0.1 mL**) and repeat.

Marks every	Instrument range	"correct reading"	Digits after the decimal point	Significant figures
10 mL				
1 mL				
0.1 mL				

a. Finish the rule in your own words: if an instrument is marked every *d* units, you may write every digit down to __________ and then estimate __________.

b. One column of your table holds steady down all three rows while the others change. Which one? Explain why that is what you should expect, given that the liquid never moved and only the marks and the vessel changed.

c. Go back to your part 1 prediction. Was answer *a* right? If not, name the specific assumption that failed.

4 Working backwards, and a shortcut that fails

The first half is what an exam question asks you to do; the second is a piece of reasoning that sounds right until you test it.

Working backwards. Each volume below was recorded correctly by somebody. On paper, state the graduation spacing their glassware must have carried, and the uncertainty in the last digit. Then set the demonstration to that instrument and spacing and confirm it accepts a reading with that many digits.

Recorded value	Marks every	Uncertain by ±	Confirmed in the demonstration?
24.7 mL			
8.42 mL			
17.4 cm			
386 mL			

The shortcut that fails. Set the instrument back to **Cylinder** with **Graduation spacing** at **1 mL**.

1. Drag **Magnification** fully **left**. Record the value beside the slider and the **In view** readout.
2. Drag **Magnification** fully **right**. Record both again.
3. Leave it fully right. Read the level, but type your answer with *one extra* decimal place, and press **Check**. Write down the headline of the response.

Magnification	Value beside the slider	"In view"	"Marks every"
fully left			
fully right			

a. Headline of the response to the extra decimal place:

b. Magnification changed something real about the picture. Say what it changed, and say why that change cannot buy you another digit. Then check your part 1(b) prediction against what happened.

The other end of the curve. Still on the 100 mL cylinder, take the **correct reading** the demonstration last drew, add the **Bottom-to-top of the curve** value to it, round to one decimal

place, type that, and press **Check**.
c. What does the demonstration say you did? Using the number in brackets after “Bottom-to-top of the curve”, state how many graduations that mistake costs you on this cylinder.

5 Solve these without the demonstration

Close the page or look away. Show your reasoning, not just the answer. Then reopen the demonstration and check yourself against it.

1. A class-A burette is marked every 0.1 mL and numbered from the top, because the reading tells you how much has been delivered. The initial reading is 3.28 mL and the final reading is 27.65 mL. Give the volume delivered, written with the correct number of digits, and state the uncertainty in the last digit of each reading.

2. A 10 mL graduated cylinder is marked every 0.1 mL. The bottom of the meniscus sits halfway between the 6.3 and 6.4 marks. Write the reading. Then say why 6.4 mL and 6.350 mL are both unacceptable records of that same level, giving a different reason for each.

3. Two people read the same water in the same 100 mL cylinder, marked every 1 mL. One records 42.6 mL, the other 42.7 mL. Decide which of them is wrong, and defend your answer in terms of which digits the marks supplied and which digit each person supplied.

4. A volume measured in a 1 L cylinder marked every 10 mL is written in a notebook as 250 mL. Explain what a reader of that notebook cannot tell from it, and rewrite the value so that they can.

> The demonstration models an ideal read: your eye level with the surface, and graduations that are powers of ten. Real glassware also carries a calibration tolerance — a class-A 100 mL cylinder is ±0.5 mL — and a temperature rating, and both are ignored here. On real data the tolerance can be larger than the digit you estimated.

6 On your own

Nothing here is graded. The demonstration holds more than this worksheet used.

- Switch to **Burette**. Read a level, then deliberately read it as though the scale counted upward from the bottom: subtract your reading from 50.00 mL and check that instead. Read what comes back, then work out for yourself why burettes are numbered the way they are.
- Compare **Bottom-to-top of the curve** across all three cylinders, as a fraction of a graduation. The bore diameter of each is named in the line under the **Magnification** slider. Which way does the meniscus depth go as the tube narrows, and does that surprise you?
- Switch to **Ruler** and read the note in the Facts panel about where the rod's lower end is sitting. If both ends of the rod had to be estimated instead of one, what would happen to the uncertainty in the length you report?